Point convergence of composite-FEG under negative comonotonicity

Abstract

The composite-FEG method converges to the Euclidean closest zero throughout its strict range of negative comonotonicity. We prove this for the original recurrence of Cai, Oikonomou and Zheng from arbitrary initial points. The fixed anchor and algorithm are unchanged. The established potential supplies the known residual rate and a weighted budget for correction errors. A shift of the actual state converts the update into a perturbed anchored iteration of a nonexpansive resolvent map. We give a self-contained nearest-point selection argument. Point convergence for other accelerated recurrences on this range is already known; the result here concerns the original method. A rational example at the boundary stepsize $1/L$ stops at a different zero, showing that nearest-zero selection need not extend to that endpoint.

Keywords: fast extragradient; comonotone inclusion; point convergence; Halpern iteration

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1 Recurrence and convergence theorem

Let $F : \mathbb{R}^d \to \mathbb{R}^d$ be globally L-Lipschitz, with $L > 0$, and let $A : \mathbb{R}^d \rightrightarrows \mathbb{R}^d$ be maximally monotone. Assume $S = F + A$ has a zero and is ρ-comonotone in the signed convention

$$\langle u - v, x - y\rangle \geq \rho \|u - v\|^2 \quad \big((x,u),(y,v) \in \operatorname{gra} S\big), \qquad -\frac{1}{2L} < \rho \leq 0. \tag{1}$$

The hypothesis concerns the sum $F + A$, not necessarily F alone. Fix

$$\max\{0, -2\rho\} < \eta < \frac{1}{L}. \tag{2}$$

Write $J_{\eta A} = (I + \eta A)^{-1}$. From arbitrary $z_0 \in \mathbb{R}^d$ and $c_0 = 0$, set $\beta_k = 1/(k+1)$, $h_k = F(z_k) + c_k$, and, for $k \geq 0$, perform

$$\begin{aligned} a_k &= \beta_k z_0 + (1 - \beta_k) z_k, \\ y_k &= a_k - (1 - \beta_k)(\eta + 2\rho) h_k, \\ z_{k+1} &= J_{\eta A}\big(a_k - \eta F(y_k) - 2\rho(1 - \beta_k) h_k\big), \\ c_{k+1} &= \frac{a_k - \eta F(y_k) - 2\rho(1 - \beta_k) h_k - z_{k+1}}{\eta}. \end{aligned} \tag{3}$$

This is composite-FEG, Algorithm 2 of Cai, Oikonomou and Zheng [1], with $y_k = z_{k+1/2}$ in their notation. The anchor is always the original z_0. The parameters ρ and η are fixed inputs. There is no assumption that $z_0 \in \operatorname{dom} A$ or $c_0 \in A(z_0)$; the resolvent implies $c_k \in A(z_k)$ for $k \geq 1$. At the first step, $\beta_0 = 1$, so

$$y_0 = z_0, \qquad z_1 = J_{\eta A}(z_0 - \eta F(z_0)). \tag{4}$$

Theorem 1. *Under* (1)–(2)*, the set* $Z = \operatorname{zer}(F + A)$ *is nonempty, closed and convex. Let* $p = P_Z z_0$ *be its unique Euclidean closest point to* z_0*. Every update in* (3) *is well-defined, and*

$$z_k \longrightarrow p, \qquad y_k \longrightarrow p, \qquad c_k \longrightarrow -F(p). \tag{5}$$

The established residual estimate is retained:

$$\|F(z_k) + c_k\| \leq \frac{2\,\|z_0 - p\|}{(\eta + 2\rho)k} \qquad (k \geq 1). \tag{6}$$

No rate for $\|z_k - p\|$ *is asserted.*

The residual rate and potential used below are due to [1]; they are not new contributions. That paper proves point convergence of this recurrence for $\rho = 0$, while its negative-comonotone analysis establishes residual convergence. The current author version, v6 of 4 December 2024, improves the constant and admits $\eta = 1/L$ in the residual theorem, but its point theorem still uses a strict upper stepsize. We use the v6 constant in (6) and keep the strict condition (2).

The broader task of constructing an accelerated method with point convergence on a negative-comonotone range is already addressed by later algorithms. In particular, Tran-Dinh and Nguyen-Trung [2, Theorem 7.3] prove point convergence for altered Nesterov coefficients; their fixed-anchor Section 4 supplies residual rates, and its subsequent point theorems change the anchor or coefficients. Our claim is about the original update (3), including its initial step. Its mechanism uses the classical nonexpansive anchored-iteration geometry of Halpern and Wittmann, as presented in [3]; we give the needed selection argument directly.

2 The existing potential and its error budget

Every finite update is defined because the resolvent of A has full domain. For $k \geq 1$ put

$$U_k = \left[\frac{k^2}{2}\left(1 + \frac{2\rho}{\eta}\right) - \frac{\rho}{\eta}k\right]\eta^2\,\|h_k\|^2 + k\eta\langle h_k, z_k - z_0\rangle. \tag{7}$$

This is the potential of [1, Section 4.1]. We reproduce its identities to fix the signs and arbitrary-initialization convention. Define

$$\delta_k = z_{k+1} - y_k, \qquad e_k = \eta\big(F(z_{k+1}) - F(y_k)\big).$$

Lemma 1. *For every* $k \geq 1$*,*

$$U_{k+1} \leq U_k - \frac{1 - \eta^2 L^2}{2}(k+1)^2\,\|\delta_k\|^2, \qquad U_1 \leq \frac{\eta^2 L^2 - 1}{2}\,\|z_1 - z_0\|^2 \leq 0. \tag{8}$$

For any $z^* \in Z$*, with* $R = \|z_0 - z^*\|$*,*

$$U_k \geq \frac{\eta(\eta + 2\rho)}{4}k^2\,\|h_k\|^2 - \frac{\eta}{\eta + 2\rho}R^2. \tag{9}$$

Consequently,

$$\|h_k\| \le \frac{2R}{(\eta+2\rho)k}, \qquad \sum_{k=1}^{\infty}(k+1)^2\,\|\delta_k\|^2 \le \frac{2\eta R^2}{(1-\eta^2L^2)(\eta+2\rho)}. \tag{10}$$

Proof. The exact identity from the recurrence, underlying [1, Lemma 4.4], is

$$\begin{aligned} U_k - U_{k+1} = \frac{(k+1)^2}{2}\big(\|\delta_k\|^2 - \|e_k\|^2\big) \\ + k(k+1)\eta\left(\langle h_{k+1}-h_k, z_{k+1}-z_k\rangle - \rho\,\|h_{k+1}-h_k\|^2\right). \end{aligned} \tag{11}$$

It can be verified by substitution into (7): the two relations needed are

$$\delta_k = \frac{k}{k+1}\eta h_k - \eta h_{k+1} + e_k, \quad z_{k+1}-z_k = \frac{z_0-z_k}{k+1} - 2\rho\frac{k}{k+1}h_k - \eta h_{k+1} + e_k.$$

The graph-pair term in (11) is nonnegative because $c_k \in A(z_k)$ and $c_{k+1} \in A(z_{k+1})$ for $k \ge 1$. Lipschitz continuity gives $\|e_k\| \le \eta L\,\|\delta_k\|$, proving the first inequality in (8).

For the initial bound let $d = z_1 - z_0$ and $D = F(z_1) - F(z_0)$. Equation (4) gives $\eta h_1 = -d + \eta D$. The coefficient in (7) at $k = 1$ is $1/2$, hence

$$U_1 = \frac{1}{2}\,\|-d+\eta D\|^2 + \langle -d+\eta D, d\rangle = \frac{\eta^2\,\|D\|^2 - \|d\|^2}{2} \le \frac{\eta^2L^2-1}{2}\,\|d\|^2.$$

This calculation does not use $c_0 \in A(z_0)$.

Comonotonicity against $(z^*, 0)$ gives $\langle h_k, z_k - z^*\rangle \ge \rho\,\|h_k\|^2$. Substituting $z_k - z_0 = (z_k - z^*) + (z^* - z_0)$ in (7) cancels its term linear in $k\rho$, and yields

$$U_k \ge \frac{\eta(\eta+2\rho)}{2}k^2\,\|h_k\|^2 + k\eta\langle h_k, z^* - z_0\rangle.$$

Young's inequality proves (9). Combining it with $U_k \le U_1 \le 0$ gives the residual bound. Telescoping (8) and using the same lower bound gives the sum in (10). ∎

In particular, since e_0 is a finite vector, (10) implies

$$\sum_{k=0}^{\infty}(k+1)^2\,\|e_k\|^2 < \infty, \qquad \sum_{k=0}^{\infty}\|e_k\| < \infty, \qquad (k+1)\,\|e_k\| \longrightarrow 0. \tag{12}$$

For the middle assertion apply Cauchy–Schwarz with the square-summable sequence $(1/(k+1))$. These are consequences of the source potential, not additional error assumptions.

3 A shifted state and nearest-zero selection

Lemma 2. *For every $v \in \mathbb{R}^d$ there is a unique pair (x, u) such that*

$$u \in S(x), \qquad v = x + \eta u.$$

The map

$$T(v) = x - 2\rho u \tag{13}$$

is nonexpansive on $\mathbb{R}^d$, and $\operatorname{Fix} T = Z$. Consequently Z is closed and convex.

Proof. The map $x \mapsto J_{\eta A}(v - \eta F(x))$ is a contraction with constant at most $\eta L < 1$. Its unique fixed point x satisfies the asserted inclusion with $u = (v - x)/\eta$. This proves full-domain existence; it is not inferred from comonotonicity alone.

For two input pairs write $\Delta x = x - x'$, $\Delta u = u - u'$. Equation (1) gives

$$\begin{aligned} \|v - v'\|^2 - \|T(v) - T(v')\|^2 &= 2(\eta + 2\rho)\langle \Delta x, \Delta u\rangle + (\eta^2 - 4\rho^2)\,\|\Delta u\|^2 \\ &\geq \eta(\eta + 2\rho)\,\|\Delta u\|^2 \geq 0. \end{aligned} \tag{14}$$

Also $v - T(v) = (\eta + 2\rho)u$, so $T(v) = v$ if and only if $u = 0$, equivalently $v \in Z$. Continuity proves that Z is closed. For completeness, if $x_1, x_2 \in \operatorname{Fix} T$, $0 \leq t \leq 1$, and $x = (1 - t)x_1 + tx_2$, nonexpansiveness gives

$$\|Tx - x_1\| \leq t\,\|x_1 - x_2\|\,, \qquad \|Tx - x_2\| \leq (1 - t)\,\|x_1 - x_2\|\,.$$

The squared-distance identity then yields

$$\|Tx - x\|^2 = (1 - t)\,\|Tx - x_1\|^2 + t\,\|Tx - x_2\|^2 - t(1 - t)\,\|x_1 - x_2\|^2 \leq 0.$$

Thus $Tx = x$, proving convexity. ∎

The map in (13) is a relaxed resolvent:

$$T = \left(1 + \frac{2\rho}{\eta}\right) J_{\eta S} - \frac{2\rho}{\eta} I.$$

Both coefficients are nonnegative and sum to one. It is used only in the proof; the algorithm evaluates $J_{\eta A}$, not $J_{\eta S}$. The nonexpansive-map interpretation and the projection geometry below belong to the classical anchored-iteration framework [3].

Proof of Theorem 1. The resolvent observations above prove well-definedness and the assertions about Z. Let $p = P_Z z_0$ and $R = \|z_0 - p\|$. The projection is characterized by

$$\langle z_0 - p, q - p\rangle \leq 0 \qquad (q \in Z). \tag{15}$$

For $k \geq 1$ define from the actual states

$$w_k = z_k - 2\rho h_k, \qquad b_k = \beta_k z_0 + (1 - \beta_k) w_k. \tag{16}$$

The definition of c_{k+1} in (3) gives the exact identities

$$z_{k+1} + \eta h_{k+1} = b_k + e_k, \qquad w_{k+1} = T(b_k + e_k). \tag{17}$$

No new trajectory or new implemented update is introduced.

Since $Tp = p$, nonexpansiveness implies

$$\|w_{k+1} - p\| \leq (1 - \beta_k)\,\|w_k - p\| + \beta_k R + \|e_k\|\,.$$

By (12),

$$M := \max\{\|w_1 - p\|, R\} + \sum_{k=1}^{\infty} \|e_k\| < \infty \tag{18}$$

is an upper bound on both $\|w_k - p\|$ and $\|b_k - p\|$, by induction. Meanwhile (10) gives $h_k \to 0$, hence

$$z_k - w_k = 2\rho h_k \longrightarrow 0, \qquad b_k - w_k = \beta_k(z_0 - w_k) \longrightarrow 0. \tag{19}$$

This proves boundedness of the original sequence, after the error budget has been established.

Every cluster point of (z_k) belongs to Z. Indeed, if $z_{k_j} \to q$, then

$$c_{k_j} = h_{k_j} - F(z_{k_j}) \longrightarrow -F(q).$$

The graph of A is closed, so $-F(q) \in A(q)$. Here graph closedness follows directly from maximal monotonicity: a limit pair satisfies the monotonicity inequality against every graph pair and must therefore belong to the graph. No closed-domain assumption is used. By (19), every cluster point of (b_k) belongs to Z as well. Boundedness, finite-dimensional compactness and (15) therefore imply

$$\limsup_{k\to\infty} \langle z_0 - p, b_k - p\rangle \le 0. \tag{20}$$

It remains to select one point rather than only the zero set. Put $s_k = \|w_k - p\|^2$. Since

$$\|b_k - p\|^2 = (1-\beta_k)^2 s_k + 2\beta_k\langle z_0 - p, b_k - p\rangle - \beta_k^2 R^2,$$

equations (17) and (18) give

$$s_{k+1} \le (1-\beta_k)s_k + \beta_k\varepsilon_k, \qquad \varepsilon_k = 2\langle z_0 - p, b_k - p\rangle + \frac{2M\|e_k\| + \|e_k\|^2}{\beta_k}. \tag{21}$$

We used $(1-\beta_k)^2 \le 1 - \beta_k$. By (12), $\|e_k\|/\beta_k \to 0$; hence (20) implies $\limsup \varepsilon_k \le 0$.

To finish without invoking an iteration-convergence theorem, fix $\epsilon > 0$ and choose $N \ge 1$ such that $\varepsilon_k \le \epsilon$ for $k \ge N$. Unroll (21), using

$$\prod_{k=N}^{n-1}(1-\beta_k) = \prod_{k=N}^{n-1}\frac{k}{k+1} = \frac{N}{n},$$

to obtain

$$0 \le s_n \le \frac{N}{n}s_N + \left(1 - \frac{N}{n}\right)\epsilon \qquad (n \ge N).$$

First let $n \to \infty$, then $\epsilon \downarrow 0$. Thus $w_k \to p$, and (19) gives $z_k \to p$. The definition of y_k, together with $h_k \to 0$ and $\beta_k \to 0$, proves $y_k \to p$. Finally $c_k = h_k - F(z_k) \to -F(p)$. The residual bound (6) is (10) with $z^* = p$.

Nothing divides by ρ or R. In particular, $\rho = 0$ is included. If $R = 0$, then (4) gives $z_1 = z_0$ and $c_1 = -F(z_0)$, so the subsequent trajectory is constant even if $c_0 = 0$ is not that graph value. ∎

4 Scope and attribution

The proof leaves the original fixed anchor, extrapolation, correction and initialization unchanged. Its additional observation is that the source's error budget controls the actual shifted recurrence (17). Residual convergence, the scalar potential, resolvent geometry, and anchored nearest-point selection each have established ancestry. The conclusion here is nearest-zero convergence for the specified composite-FEG map at negative ρ.

The next elementary example explains why the nearest-zero conclusion cannot simply include the upper endpoint. It does not give nonconvergence to the zero set.

Example 1. Let $L = \eta = 1$, $z_0 = (0,1)^T$, $c_0 = 0$, and take any $\rho \in (-1/2, 0]$. Set $F(x) = Mx$ and $A(x) = Hx$, where

$$M = \begin{pmatrix} -1/5 & 3/5 \\ 3/5 & 7/10 \end{pmatrix}, \qquad H = \begin{pmatrix} 1/5 & -3/5 \\ -3/5 & 23/10 \end{pmatrix}. \tag{22}$$

Then the exact orbit of (3) satisfies, for every $k \geq 1$,

$$z_k = q := \begin{pmatrix} -1/2 \\ 0 \end{pmatrix}, \qquad c_k = \begin{pmatrix} -1/10 \\ 3/10 \end{pmatrix}, \qquad h_k = 0. \tag{23}$$

The zero set is $Z = \mathbb{R} \times \{0\}$, but $q \neq P_Z z_0 = (0,0)^T$.

Proof. The symmetric matrix M has eigenvalues 1 and $-1/2$, so its norm is exactly one. The leading principal minor of H is $1/5$ and its determinant is $1/10$, so H is positive definite. Moreover, $M + H = \operatorname{diag}(0, 3)$ is monotone and thus satisfies (1) for every specified ρ. These are the full source assumptions; F itself need not be monotone.

Exact multiplication gives

$$(I+H)^{-1} = \begin{pmatrix} 11/12 & 1/6 \\ 1/6 & 1/3 \end{pmatrix}, \qquad C := (I+H)^{-1}(I-M) = \begin{pmatrix} 1 & -1/2 \\ 0 & 0 \end{pmatrix}.$$

The prescribed first step gives $z_1 = Cz_0 = q$, $c_1 = Hq = (-1/10, 3/10)^T$, and $h_1 = 0$. If (23) holds at k, the predictor is

$$y_k = \beta_k z_0 + (1-\beta_k)q = \begin{pmatrix} -k/[2(k+1)] \\ 1/(k+1) \end{pmatrix}.$$

The next correction gives $z_{k+1} = Cy_k = q$ because $Cz_0 = Cq = q$, and $c_{k+1} = Hq$. This closes the induction, uniformly in ρ. ∎

Here the points converge and the residual is zero from index one onward. Only Euclidean nearest-zero selection fails. The loss of the strict proof's error budget is also explicit: since $M(1/2, 1)^T = (1/2, 1)^T$,

$$\delta_k = e_k = -\frac{1}{k+1}\begin{pmatrix} 1/2 \\ 1 \end{pmatrix}, \qquad (k+1)^2 \|\delta_k\|^2 = \frac{5}{4} \quad (k \geq 1).$$

Thus the weighted error sum diverges despite the stationary point sequence. Convergence to some zero for the full nonlinear endpoint class is not settled by this example.

The strict upper bound $\eta L < 1$ is used both to construct the full-domain comparison resolvent by contraction and to obtain a positive coefficient in (8). We do not extend the point theorem to $\eta = 1/L$, despite the current source residual theorem including that endpoint. Likewise $\eta + 2\rho > 0$ is essential in the residual bounds and fixed-point identification. The model uses all graph pairs in (1); a weak-Minty inequality only against a zero does not supply the argument. No point-distance rate, infinite-dimensional extension, or new general acceleration principle is asserted.

References

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